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Complexity Adjusted Comparisons of LPs and VARs: Evidence from Monte-Carlo Simulations under Functional Misspecification

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Abstract

Vector autoregressions (VARs) and local projections (LPs) estimate the same population impulse responses, yet they are usually compared at a common lag order, which confounds the choice of estimator with the choice of model complexity. We revisit this comparison in a Monte Carlo study that holds effective complexity constant, following the complexity-adjusted benchmark of Ludwig (2026). Under correct specification, a fixed LP(4) already behaves like a highly parameterized VAR of order $q \approx 15$, so the familiar point-estimation edge of low-order VARs reflects parsimony rather than an intrinsic advantage of the iterated method. We then introduce functional misspecification through a nonlinear lag-quadratic process. Because the centered quadratic term is nearly orthogonal to the linear lag space, the target impulse response stays almost unchanged, so the equal-lag VAR(4) keeps its root mean squared error advantage over LP(4) and the finite-sample gap is driven by variance alone. The nonlinearity instead surfaces as conditional heteroskedasticity in the residuals, which the homoskedastic VAR delta-method intervals cannot accommodate. They undercover ever more severely as persistence and nonlinearity rise, while the heteroskedasticity-robust intervals of the local projection stay near nominal. Population equivalence of the two estimands therefore does not carry over to finite-sample inference, and the better method depends on whether the researcher prioritizes the efficiency of the VAR point estimate or the interval validity of the direct projection.

Code and replication materials
github.com/Aaron030/Monte_Carlo-LP_vs_VAR.

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1. Introduction

Impulse responses to exogenous shocks are estimated through two main approaches, vector autoregressive (VAR) and Local Projection (LP) models. Since the introduction of the latter, an extensive literature has compared the two. One strand establishes that, under relatively general conditions, LPs and VARs estimate the same population impulse responses (Fusari et al., 2026; Plagborg-Møller & Wolf, 2021). In finite samples at equal lag lengths, however, LPs were often viewed as less sensitive to misspecification (Jordà, 2005; Ramey, 2016), a property that Montiel Olea et al. (2024) subsequently formalise, whereas VARs were seen as more statistically efficient (Li et al., 2024; Montiel Olea et al., 2025). Ludwig (2026) explains this finite-sample bias-variance tradeoff analytically, showing that an LP can be represented as a sequence of VAR estimations of increasing lag length as the horizon grows, and therefore argues for comparing LPs and VARs by effective complexity rather than lag length.

Our paper asks two questions. First, under correct specification, where does a fixed LP estimator sit relative to the full complexity-adjusted VAR frontier, and does the conventional equal-lag ranking survive once LP(4) is compared against the entire VAR-order frontier rather than only the equal-lag VAR(4)? To answer this, we benchmark a fixed LP(4) against progressively higher-order VAR specifications (De Graeve & Westermark, 2025) across empirically relevant sample sizes (see e.g. Herbst & Johannsen, 2024). Second, once controlled functional misspecification is introduced through a nonlinear DGP, does any genuine LP advantage emerge, and does it manifest in point estimation, in inference, or both? Since Plagborg-Møller and Wolf (2021) establish that LP and VAR target the same best linear approximation in population regardless of whether the DGP is linear, we augment the baseline VAR(4) with a mean-zero quadratic term in lagged state variables and examine how LP(4), the equal-lag VAR(4), and higher-order VARs respond as the nonlinearity strengthens.

Our findings suggest that under correct specification the conventional point-estimation advantage of equal-lag VARs over the LP persists, but it is largely an artifact of mismatched model complexity. A fixed LP(4) already operates like a highly parameterized system, sitting high on the complexity-adjusted VAR frontier (beyond $q \approx 15$), while that parametric efficiency leaves VAR inference fragile and sharply under-covering near the unit-root boundary, where local projections retain near-nominal coverage. Introducing functional misspecification through a nonlinear lag-quadratic DGP leaves the shared best linear estimand almost unchanged, so the equal-lag VAR(4) keeps its mean squared error edge over LP(4) and the finite-sample gap is driven purely by variance. The nonlinearity instead surfaces as conditional heteroskedasticity in the residuals, which homoskedastic VAR delta-method intervals cannot accommodate but LP's heteroskedasticity-robust intervals can. Local projections therefore deliver the only valid inference, the first setting in this study where their robustness yields a tangible advantage.

The remainder of the paper is structured as follows. Section 2 reviews vector autoregressions and local projections, the empirical literature on their bias-variance trade-off at equal lag lengths, and their asymptotic and finite-sample equivalence. Section 3 presents the baseline simulation study under correct specification, Section 4 introduces functional misspecification through a nonlinear DGP, and Section 5 concludes.

2. Theoretical Background

An impulse response measures how an outcome reacts over time to a single shock. Following Koop et al. (1996), it can be defined nonparametrically as the difference between two forecasts of y_{t+h}. Both forecasts use the same controls $\mathbf{x}_t$ and differ only in the time-t value of the variable of interest s_t. One forecast sets $s_t = s_0+\delta$, raising the baseline s_0 by a shock of size δ, while the other is kept at s_0. The controls $\mathbf{x}_t$ hold fixed the information used to identify the response, namely the lags of the system and, under some identification schemes, contemporaneous variables. Following Ramey (2016), a shock is an exogenous and unanticipated movement that is uncorrelated with the controls in $\mathbf{x}_t$ and with all other structural shocks. At horizon h, the gap between the two forecast paths is the causal effect of the shock in population,

$$IR(h, \delta) \equiv \mathbb{E}[\, y_{t+h} \mid s_t = s_0 + \delta;\, \mathbf{x}_t] - \mathbb{E}[\, y_{t+h} \mid s_t = s_0;\, \mathbf{x}_t] \,. \tag{1}$$

Vector autoregressions and local projections are the two standard ways to estimate this object in finite samples. A VAR, introduced by Sims (1980), fits the joint dynamics of all variables as one autoregressive system and reads impulse responses off the estimated model by iterating it forward. A local projection, introduced by Jordà (2005), instead estimates the response at each horizon with its own OLS regression of the future outcome on current and past data. Because they build the response differently, the two estimators accumulate estimation error in different ways across horizons. At an equal lag order they sit at opposite ends of a bias-variance spectrum, with local projections carrying more variance and VARs more bias under misspecification (Li et al., 2024; Montiel Olea et al., 2025). Ludwig (2026) argues that this equal-order comparison is misleading, because at a given lag order p an LP(p) is strictly more flexible than a VAR(p). The usual bias-variance trade-off therefore mixes the choice of estimator with the choice of model complexity.

2.1 Introduction to VARS

Following Lütkepohl (2005), we define a VAR(p) population model in which the outcome of interest y_t is a $K \times 1$ random vector, the A_i are fixed $K \times K$ coefficient matrices, and u_t is K-dimensional white noise with $\mathbb{E}[u_t] = 0$, a nonsingular covariance matrix $\mathbb{E}[u_t u_t'] = \Sigma_u$, and $\mathbb{E}[u_t u_s'] = 0$ for $t \neq s$. Rather than write out all p lags, we cast the VAR(p) in its equivalent VAR(1) companion form,

$$Y_t = \nu + \mathbf{A} Y_{t-1} + U_t \tag{2}$$

where

$$\nu := \underset{Kp\times 1}{\begin{bmatrix} \nu \\ 0 \\ \vdots \\ 0 \end{bmatrix}}, \quad Y_t := \underset{Kp\times 1}{\begin{bmatrix} y_t \\ y_{t-1} \\ \vdots \\ y_{t-p+1} \end{bmatrix}}, \quad \mathbf{A} := \underset{Kp\times Kp}{\begin{bmatrix} A_1 & A_2 & \cdots & A_{p-1} & A_p \\ I_K & 0 & \cdots & 0 & 0 \\ 0 & I_K & & 0 & 0 \\ \vdots & & \ddots & & \vdots \\ 0 & 0 & \cdots & I_K & 0 \end{bmatrix}}, \quad U_t := \underset{Kp\times 1}{\begin{bmatrix} u_t \\ 0 \\ \vdots \\ 0 \end{bmatrix}}.$$

If Y_t satisfies the stability condition, so that all roots z of the characteristic equation lie strictly outside the unit circle, the process is covariance stationary and admits a Wold moving-average (MA) representation

$$\det(I_{Kp} - \mathbf{A}z) \neq 0 \text{ for } |z| \leq 1 \iff Y_t = \mu + \sum_{\ell=0}^{\infty} \mathbf{A}^l U_{t-\ell}, \qquad \text{where } \mu := \mathbb{E}(Y_t) \tag{3}$$

Keeping only the top K rows gives the moving-average representation for y_t

$$y_t = \mu + \sum_{\ell=0}^{\infty} \Psi_\ell \, u_{t-\ell}, \qquad \Psi_\ell := [\mathbf{A}^\ell]_{1:K,\,1:K} \tag{4}$$

where Ψ_h is the reduced-form impulse response at horizon h, the response of y_{t+h} to a reduced-form innovation u_t. It is the upper-left $K \times K$ block of $\mathbf{A}^h$, because $U_{t-\ell} = (u_{t-\ell}, 0, \ldots, 0)'$ is nonzero only in its first K positions. The components of u_t are generally correlated at the same date, $\Sigma_u = \mathbb{E}(u_t u_t') \neq I_K$, so a shock to one component moves the others contemporaneously. The reduced-form responses Ψ_h therefore carry no structural interpretation. To identify structural shocks, we restrict the impact matrix B in $u_t = B\varepsilon_t$, where $\varepsilon_t \sim (0, I_K)$ are mutually uncorrelated structural shocks and $\Sigma_u = BB'$.

The most common choice is recursive Cholesky identification (Ramey, 2016), which makes B lower triangular with positive diagonal entries and so recovers it uniquely as the Cholesky factor of Σ_u. The ordering imposes a causal structure, as the variable ordered first responds contemporaneously to all shocks while the variable ordered last is insulated from all contemporaneous feedback. The diagonal entries b_{ii}, $i = 1, \ldots, K$, measure the direct impact of structural shock i on variable i, and the off-diagonal entries b_{ij}, $1 \leq j < i \leq K$, capture the contemporaneous spillover from shock j to variable i. Substituting $u_t = B\varepsilon_t$ into (4) yields the structural moving-average representation

$$y_t = \mu + \sum_{\ell=0}^{\infty} \Theta_\ell \, \varepsilon_{t-\ell}, \qquad \Theta_\ell := \Psi_\ell B, \qquad B = \begin{pmatrix} b_{11} & 0 & \cdots & 0 \\ b_{21} & b_{22} & \cdots & 0 \\ \vdots & & \ddots & \vdots \\ b_{K1} & b_{K2} & \cdots & b_{KK} \end{pmatrix} \tag{5}$$

where the (i,j)-th element of Θ_h is the response of variable i to a unit innovation in shock j at horizon h. The structural impulse response of variable i to shock j is thus

$$\theta_h^{ij} = [\Psi_h B]_{ij} \tag{6}$$

where $[\,\cdot\,]_{ij}$ denotes the (i,j)-th element of a matrix. Leading (5) forward by h periods gives

$$y_{t+h} = \mu + \sum_{\ell=0}^{\infty} \Theta_\ell \, \varepsilon_{t+h-\ell}. \tag{7}$$

Because the structural shocks are mutually and serially uncorrelated, with $\varepsilon_t \sim (0, I_K)$, projecting y_{t+h} on the date-t shock ε_t keeps only the $\ell = h$ term, since all leads and lags are orthogonal to ε_t. The response of variable i to a unit innovation in shock j at horizon h is therefore

$$\theta_h^{ij} = \frac{\partial\, \mathbb{E}[y_{i,t+h} \mid \varepsilon_{j,t}]}{\partial\, \varepsilon_{j,t}} = [\Theta_h]_{ij} = [\Psi_h B]_{ij}, \tag{8}$$

which is exactly the population object $\mathrm{IR}(h, \delta)$ of Equation (1), specialised to a unit structural innovation in shock j.

Estimation of the structural impulse response $\hat{\theta}_h^{ij}$ proceeds in two stages. In the first, the stacked coefficient matrix $A := [A_1, \ldots, A_p]$ of dimension $K \times Kp$ is estimated equation by equation via OLS. If the process satisfies the stability condition and the innovations u_t form a strictly stationary martingale difference sequence with positive definite covariance and finite fourth moments, this estimator is asymptotically normal (Hansen, 2022),

$$\sqrt{T}\left(\mathrm{vec}(\hat{A}) - \mathrm{vec}(A)\right) \xrightarrow{d} \mathcal{N}(0,\ V_\alpha), \tag{9}$$

where the asymptotic covariance takes the robust sandwich form $V_\alpha = \overline{\Gamma}^{-1}\Omega\overline{\Gamma}^{-1}$. Here $\overline{\Gamma} = I_K \otimes \Gamma$, with $\Gamma = \mathbb{E}(Z_t Z_t')$ the second-moment matrix of the stacked regressors $Z_t = (1, y_{t-1}', \ldots, y_{t-p}')'$, and $\Omega = \mathbb{E}(u_t u_t' \otimes Z_t Z_t')$. Under conditional homoskedasticity, Ω simplifies to $\Sigma_u \otimes \Gamma$ and the asymptotic variance reduces to the standard form $\Gamma^{-1} \otimes \Sigma_u$ (Lütkepohl, 2005), where OLS is asymptotically efficient. The general form V_α remains necessary for valid inference when the data-generating process is conditionally heteroskedastic.

In the second stage, the structural objects are recovered from the reduced-form estimates. The impact matrix $\hat{B}$ is the Cholesky factor of $\hat{\Sigma}_u$, and the reduced-form impulse responses follow from the recursion

$$\hat{\Psi}_h = \sum_{\ell=1}^{\min(h,p)} \hat{A}_\ell\, \hat{\Psi}_{h-\ell}, \qquad \hat{\Psi}_0 = I_K. \tag{10}$$

so that the structural impulse response estimator $\hat{\theta}_h^{ij} = g(\hat{\alpha}, \hat{\sigma})$, with $\alpha = \mathrm{vec}(A)$ and $\sigma = \mathrm{vec}(\Sigma_u)$, is a smooth nonlinear function of the first-stage estimates,

$$\hat{\theta}_h^{ij} = \left[\hat{\Psi}_h\, \hat{B}\right]_{ij}. \tag{11}$$

Since g is differentiable in $(\hat{\alpha}, \hat{\sigma})$, the delta method gives

$$\sqrt{T}\left(\hat{\theta}_h^{ij} - \theta_h^{ij}\right) \xrightarrow{d} \mathcal{N}\big(0,\ \nabla g(\alpha,\sigma)'\, \Sigma_{\alpha,\sigma}\, \nabla g(\alpha,\sigma)\big), \tag{12}$$

where $\Sigma_{\alpha,\sigma}$ is the joint asymptotic covariance of $(\hat{\alpha}', \hat{\sigma}')'$ and the Jacobian $\nabla g(\alpha, \sigma)$ has a closed-form expression (Lütkepohl, 1990). These confidence intervals are valid pointwise in h under stationarity, but not uniformly over the persistence of the process or over growing horizons, which motivates a direct look at finite-sample behaviour.

These favourable asymptotic properties are qualified by three related problems at the sample sizes typical of macroeconomic work. First, estimation error propagates through iteration. Because $\hat{\Psi}_h$ is the upper-left block of $\hat{\mathbf{A}}^h$, error in $\hat{A}$ enters the impulse response through powers of the companion matrix and compounds as the horizon grows. Even small coefficient biases then translate into large distortions at longer horizons (Kilian, 1998). With unit or near-unit roots, Phillips (1998) shows that long-horizon responses from unrestricted VARs are not even consistently estimated, converging to random variables rather than the true responses.

Second, the autoregressive coefficients are biased downward in short samples, as least squares underestimates the persistence of the process (Marriott & Pope, 1954; Nicholls & Pope, 1988; Pope, 1990). Because $\hat{\Psi}_h$ inherits this bias through the recursion, the estimated responses decay too quickly and understate the true persistence. The distortion is worst when persistence is high and the sample is short.

Third, these distortions undermine delta-method inference. As the largest roots approach the unit circle, the normal approximation deteriorates and the confidence intervals undercover, increasingly so at longer horizons (Kilian, 1998; Kilian, 1999).

A further class of distortions arises when the autoregressive structure is itself misspecified. If the true process has moving-average components, a finite-order VAR is generically inconsistent regardless of sample size, and although higher-order approximations can asymptotically absorb the omitted MA dynamics, the required lag expansion quickly exhausts degrees of freedom in samples of typical macroeconomic length (Braun & Mittnik, 1993). The consequences of lag choice are also asymmetric. Overfitting only inflates variance, whereas underfitting omits higher-order dynamics of the response and produces spuriously tight intervals with coverage far below nominal (Kilian, 2001). Both channels are population-level phenomena. Ravenna (2007) shows that truncation bias in the VAR coefficients, and the resulting contamination of the structural rotation matrix, persists even as $T \to \infty$.

2.2 Introduction to Local Projections

Whereas the VAR recovers the impulse response by iterating a single fitted model forward, the local projection estimates each horizon directly from its own regression (Jordà, 2005). Following Jordà and Taylor (2025), the LP obtains the horizon-h response of the outcome y to the shock s_t by projecting the future outcome y_{t+h} onto s_t while controlling for the conditioning set $\mathbf{x}_t$ that spans the information of the system,

$$y_{t+h} = \alpha_h + \beta_h\, s_t + \boldsymbol{\gamma}_h'\, \mathbf{x}_t + \xi_{t+h}^{(h)}, \qquad h = 0, 1, \ldots, H, \tag{13}$$

where y_{t+h} is the scalar response and s_t the scalar shock of interest, which for an observed structural innovation is $s_t = \varepsilon_{j,t}$. The control vector $\mathbf{x}_t = (y_{t-1}', \ldots, y_{t-p}')'$ collects the p lags that span the conditioning information and, under non-recursive identification, may also include contemporaneous controls. Here α_h is a constant that can be omitted without loss of generality, $\boldsymbol{\gamma}_h$ a $(Kp \times 1)$ vector of coefficients on the controls, and $\xi_{t+h}^{(h)}$ the horizon-h projection residual. The leading coefficient β_h is the impulse response itself. When y is variable i and $s_t = \varepsilon_{j,t}$, it coincides with the structural response θ_h^{ij} of Equation (6) and with the population object $\mathrm{IR}(h, \delta = 1)$ of Equation (1).

Under appropriate identification, in the simplest case exogeneity of s_t conditional on $\mathbf{x}_t$, the impulse response is the OLS coefficient $\hat{\beta}_h$ on s_t in Equation (13). The LP thus recovers the response without parametrizing the propagation dynamics, unlike the VAR, which relies on a fitted autoregressive model and forward iteration. When conditional exogeneity fails, causal interpretation can instead be restored by enriching $\mathbf{x}_t$ to achieve selection on observables (Montiel Olea et al., 2025), by instrumenting s_t with an external proxy under a relevance and lead-lag exogeneity condition (Jordà et al., 2015; Stock & Watson, 2018), or by importing a conventional VAR identification scheme such as the recursive Cholesky ordering (Plagborg-Møller & Wolf, 2021). The instrumental-variable route is moreover robust to non-invertibility, recovering the response even when the structural shock is not spanned by current and past observables and the recursive scheme therefore fails.

Unlike the VAR, estimation of $\hat{\beta}_h$ proceeds in a single stage, directly by OLS on Equation (13), whose leading coefficient is the horizon-h impulse response (Jordà & Taylor, 2025). Because $\hat{\beta}_h$ is the coefficient on a regressor that is orthogonal to the projection error $\xi_{t+h}^{(h)}$ by construction of the structural shock, it recovers the response without passing through a nonlinear mapping, and hence without the delta-method step that governs the VAR estimator in Equation (12).

Provided that the process $\{y_t\}$ is strictly stationary and ergodic, with a positive definite covariance matrix, bounded moments $\mathbb{E}\|y_t\|^r < \infty$ for some $r > 4$, and mixing coefficients satisfying $\sum_{\ell=1}^{\infty} \alpha(\ell)^{1-4/r} < \infty$, the least-squares estimator is asymptotically normal (Hansen, 2022). Letting $n = T - h$ denote the effective sample size at horizon h and $\mathbf{w}_t = (s_t, \mathbf{x}_t')'$ the stacked vector of all regressors, we have

$$\sqrt{n}\left(\hat{\mathbf{b}}_h - \mathbf{b}_h\right) \xrightarrow{d} \mathcal{N}(\mathbf{0},\ V_h)\,, \tag{14}$$

where $\mathbf{b}_h = (\beta_h, \boldsymbol{\gamma}_h')'$ is the vector of all projection coefficients and $\hat{\mathbf{b}}_h$ its OLS estimate. The asymptotic covariance matrix takes the robust sandwich form

$$V_h = Q^{-1}\Omega_h Q^{-1}, \tag{15}$$

where $Q = \mathbb{E}(\mathbf{w}_t\mathbf{w}_t')$ is the second-moment matrix of the regressors, and Ω_h is the long-run variance matrix of the projection scores, defined as

$$\Omega_h = \sum_{\ell=-\infty}^{\infty} \mathbb{E}\Big[\big(\xi_{t+h-\ell}^{(h)}\mathbf{w}_{t-\ell}\big)\big(\xi_{t+h}^{(h)}\mathbf{w}_t\big)'\Big]. \tag{16}$$

The long-run form of this covariance matrix reflects a basic feature of direct projection. Because the predictive regression aggregates the shocks accruing between t and $t + h$ over overlapping forecast windows, the projection error $\xi_{t+h}^{(h)}$ follows a moving-average process of order $h - 1$ and is therefore serially correlated.

To illustrate this mechanism, consider the stylized AR(1) DGP of Jordà and Taylor (2025), with persistence parameter ρ, initial condition $y_0 = 0$, and a strictly stationary error term u_t satisfying $\mathbb{E}[u_t|\{u_s\}_{s\neq t}] = 0$ almost surely

$$y_t = \rho y_{t-1} + u_t. \tag{17}$$

With the simplified local projection of Equation (13), the h-step regression is

$$y_{t+h} = \beta_h\, y_t + \xi_{t+h}^{(h)} \tag{18}$$

and recursive substitution of the AR(1) law of motion into (18), writing $\beta(\rho, h)$ for the LP IRF estimator of ρ^h and $\xi(\rho, h)$ for the corresponding residual, yields

$$y_{t+h} = \beta(\rho, h) y_t + \xi(\rho, h), \qquad \xi(\rho, h) = \sum_{j=1}^{h} \rho^{h-j} u_{t+j}. \tag{19}$$

The stronger the persistence ρ, the larger and more slowly decaying the autocovariances of $\xi(\rho, h)$, which worsens the distortion in naive OLS standard errors. As $\rho \to 1$, OLS also exhibits non-standard near-unit-root behaviour, invalidating inference based on standard normal critical values. As long as ρ lies safely inside the unit circle ($|\rho| < 1$), however, the LP estimator is consistent, $\hat{\beta}_h \xrightarrow{p} \beta_h = \theta_h^{ij}$, and remains asymptotically normal once the serially correlated residuals are appropriately corrected (Jordà & Taylor, 2025; Montiel Olea & Plagborg-Møller, 2021).

To address this serial correlation, Jordà (2005) originally proposed Newey-West type (Newey & West, 1987) heteroskedasticity- and autocorrelation-consistent (HAC) standard errors. In practice, HAC inference brings its own complications. It requires tuning parameters with complex interpretations (Lazarus et al., 2018) and can itself be a source of finite-sample bias. Herbst and Johannsen (2024) show that, although HAC estimators are asymptotically valid, the long-run variance is typically downward-biased in short samples, which systematically understates the uncertainty around the point estimate.

To allow valid inference even near unity, Montiel Olea and Plagborg-Møller (2021) augment the local projection with one additional lag. Under the weak assumption that the shock is unpredictable from its own past and future, this makes the regression score, the product of the projection residual $\xi_{t+h}^{(h)}$ and the regressor of interest after partialling out the controls, serially uncorrelated. Conventional Eicker–Huber–White (EHW) heteroskedasticity-robust standard errors then suffice despite the serially correlated residual. The resulting t-statistic is asymptotically standard normal uniformly over $\rho \in [-1, 1]$ and horizons $h/T \to 0$.

A long-standing piece of conventional wisdom holds that local projections are more robust to misspecification than VARs, originating with Jordà (2005, p. 162) and echoed across later surveys (Li et al., 2024; Ramey, 2016). Plagborg-Møller and Wolf (2021) document this view while noting that it had long lacked a formal foundation. In their finite-order treatment, LPs are indeed generally more robust to misspecification than equal-order VARs, a property they establish without invoking the population equivalence of the two estimands. Montiel Olea et al. (2024) show why. Because the LP reads each horizon off a separate regression rather than iterating a single fitted model forward, a misspecification of the short-run dynamics does not compound through the powers of the companion matrix as it does for the VAR, whose bias arises precisely from extrapolating the response through the iterated mapping $\hat{A}^h$ that the misspecified model no longer satisfies.

As with VARs, however, Herbst and Johannsen (2024) show that LP point estimates are biased downward in finite samples, for two reasons. First, and unlike VARs, the finite-sample estimation of the intercept distorts the sample covariance between shock and outcome even when the two are uncorrelated in population. Second, and as with VARs, the estimation of the lagged-variable coefficients inherits the classical downward bias of least-squares autoregressive estimators (Pope, 1990) and grows the more persistent the controls are. Both channels make the bias depend on the true impulse responses at neighbouring horizons and worsen with persistence and the horizon, as it accumulates over an expanding sum of terms as h increases. Although Herbst and Johannsen (2024) propose a bias-corrected estimator, Li et al. (2024) find that it reduces bias only partially and at the cost of higher variance.

2.3 On the Equivalence of VARs and Local Projections

As local projection inference gained popularity over the last decade, a growing body of simulation evidence has compared the finite-sample performance of LP and VAR estimators across data-generating processes of differing richness. Two findings recur. When the DGP admits an exact finite-order VAR representation, the VAR is correctly specified at the true lag order, dominates LP on mean-squared-error criteria at intermediate and long horizons, and is thus the preferable choice for point estimation (Li et al., 2024; Montiel Olea et al., 2024). On coverage, Kilian and Kim (2011) and Brugnolini (2018) show that LP intervals are less accurate than an equivalent VAR benchmark. Although this largely reflects the standard error chosen for the local projection, Montiel Olea et al. (2024) find no LP advantage even when switching from HAC to EHW standard errors. The added flexibility of least-squares LP therefore yields little bias reduction over the already well-specified VAR, while carrying substantially higher variance.

When the DGP departs from any finite-order VAR, however, as in the empirically calibrated dynamic factor models of Li et al. (2024) or under an extension to VARMA processes, a genuine bias-variance trade-off emerges. Here the least-squares LP has lower bias than the VAR but substantially higher variance at intermediate and long horizons, so VAR methods remain preferable under MSE unless researchers place a higher weight on bias reduction (Li et al., 2024). On coverage, Montiel Olea et al. (2024) show that LPs with EHW inference yield correctly centred intervals, whereas VARs severely undercover and their bias leaves the intervals off-centre. For lag truncation, Brugnolini (2018) varies the common lag length of LPs and VARs against a true VAR(12) process and shows that LP coverage improves relative to VARs as the gap to the true lag length widens.

All these simulation studies, however, compare LPs and VARs at a single common lag length p, a design that presupposes the two methods estimate the same object. Plagborg-Møller and Wolf (2021) establish that, under weak stationarity, an unrestricted lag structure, and a common identification scheme, the LP and VAR estimators converge to the identical structural impulse response function in population, up to a constant of proportionality. Any conventional LP impulse response can therefore be obtained from an appropriately ordered recursive VAR, and vice versa. Moreover, because both methods act as best linear predictors of the data-generating process, the authors show that they are symmetrically robust to nonlinear misspecification in population.

Crucially for the equal-lag designs above, Plagborg-Møller and Wolf (2021) further show that when both specifications are fixed at order p, their population estimands agree exactly up to horizon $h \leq p$ but generally diverge for $h > p$. The divergence is thus localized to horizons beyond the estimation lag length, precisely the intermediate and long horizons where the simulation literature finds the bias-variance trade-off most pronounced. In practice, then, LP versus VAR is not a choice between different population objects but between two finite-sample estimators of the same structural response, distinguished mainly by their bias and variance.

While Plagborg-Møller and Wolf (2021) locate the divergence at horizons $h > p$, they do not characterize how local projections actually construct these longer-horizon responses in finite samples. Ludwig (2026) closes this gap with an exact least-squares identity that holds not only in population but in any finite sample. The h-step impulse response of an LP(p) equals the forward iteration of h one-step predictions from a sequence of VARs of increasing order,

$$\mathrm{IRF}_h^{\mathrm{LP}(p)}(d) = \begin{bmatrix} I_K & \mathbf{0} \end{bmatrix} \Pi(p+h-1) \cdots \Pi(p+1)\, \Pi(p) \begin{bmatrix} d \\ \mathbf{0} \end{bmatrix}, \tag{20}$$

where $\Pi(q)$ is the estimated companion matrix of a VAR(q), I_K is the identity matrix matching the dimension of the system, and d is the initial structural shock vector.

At $h = 1$ the two methods coincide. At $h = 2$, however, the LP(p) is no longer the simple two-step iterate of a VAR(p). Instead it advances the one-step forecast with a VAR($p + 1$), and each further horizon effectively adds one lag to the implicit VAR. This identity has profound implications for model comparison. At equal order, LPs and VARs are neither equally parsimonious nor cast in the same recursive language. An LP(p) at horizon h is built from VARs of orders $p, p + 1, \ldots, p + h - 1$, whereas a standard VAR(p) holds the order fixed at every step.

The fixed-order VAR(p) is therefore a heavily restricted special case of the equal-order LP(p), which is itself a restricted special case of a highly parameterized VAR($p{+}h{-}1$). Seen this way, the familiar pattern of lower bias but higher variance for LPs relative to a same-order VAR is largely a mechanical consequence of comparing a more flexible, over-parameterized specification with its own restriction, rather than an intrinsic property of the projection method. This confounds the standard reading of the bias-variance trade-off, because at equal lag orders the methodological effect (LP vs. VAR) cannot be cleanly separated from the complexity effect (more vs. fewer effective lags).

3. Baseline Simulation

Because Ludwig (2026)'s contribution is analytical and deliberately abstains from a simulation-based horse race, whether the conventional finite-sample conclusions survive once the equal-lag restriction is relaxed remains open. We therefore operationalize his central implication in a controlled, correctly specified setting. Rather than comparing a fixed LP(4) to a single equal-lag VAR(4), we locate it on the full VAR-order frontier from VAR(1) to VAR($p + H - 1$), rendering the implicit complexity of the direct projection visible as a position on that frontier. Because the identity in Equation (20) maps LP(p) to a sequence of VARs rather than to any single higher-order VAR, no individual VAR is its perfect counterpart, so the LP(4) is read as a reference line against the frontier, against which its bias, variance, and coverage are located. Beyond operationalizing the complexity argument, this correctly specified design also serves as the reference against which the later extensions measure the effect of introducing misspecification.

3.1 Simulation Design

The underlying DGP is a stable bivariate zero-mean VAR(4) with recursive Cholesky identification. Consistent with the theoretical setup, this imposes a contemporaneous zero restriction where the first variable does not respond on impact to shocks in the second variable, while the second variable may respond to both shocks simultaneously.

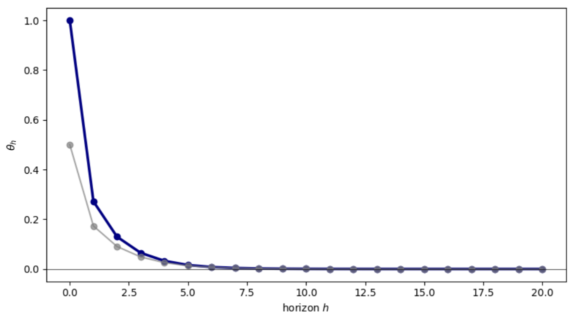
$$y_t = A_1 y_{t-1} + A_2 y_{t-2} + A_3 y_{t-3} + A_4 y_{t-4} + u_t, \quad y_t \in \mathbb{R}^2 \tag{21}$$

$$\Sigma_u = BB', \qquad B = \begin{pmatrix} 1 & 0 \\ 0.1 & 0.5 \end{pmatrix}, \quad b_{11}, b_{22} > 0 \tag{22}$$

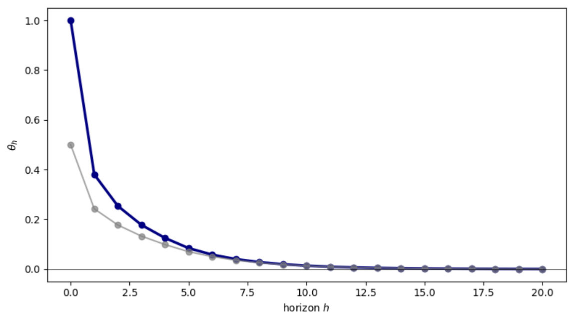
Persistence is calibrated via the spectral radius of the companion matrix $\mathcal{A}$ across the empirically relevant range $\rho \in \{0.5, 0.7, 0.95\}$. The lower bound reflects weakly persistent processes where impulse responses decay rapidly, rendering bias-variance comparisons at longer horizons uninformative. The upper bound approaches the unit-root boundary while maintaining covariance stationarity, covering the high-persistence region where finite-sample differences between estimators are most pronounced.

Figure 1: True Impulse Response Functions Across Persistence Regimes

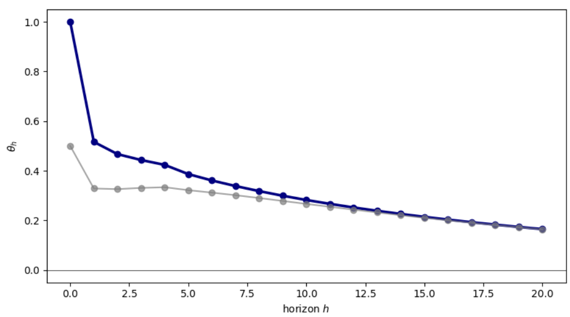
(a) $\rho = 0.5$



(b) $\rho = 0.7$



(c) $\rho = 0.95$



Note. Each panel shows the true structural impulse responses of variables one and two to a one-standard-deviation shock in the first structural innovation, traced over horizons $h = 1, \ldots, 20$. Panels correspond to the persistence regimes $\rho \in \{0.5, 0.7, 0.95\}$. The blue line is the response of y_1 to ε_1, which is the estimand θ_h, and the grey line is the response of y_2 to ε_1.

The structural target of interest is the impulse response function, which traces the dynamic causal effect of a structural innovation on the system over time. We normalize the innovations as

$u_t = B\varepsilon_t$ with $\varepsilon_t \overset{\text{iid}}{\sim} \mathcal{N}(\mathbf{0}, I_2)$, so that a unit innovation in $\varepsilon_{1,t}$ is by construction a one-standard-deviation shock. The population estimand at horizon h is then

$$\theta_h = \Psi_h B e_1, \qquad e_1 = (1,0)', \qquad h = 0, 1, \ldots, 20, \tag{23}$$

where Ψ_h is the reduced-form moving-average matrix generated by the companion recursion $\Psi_h = \sum_{j=1}^{\min(h,4)} A_j \Psi_{h-j}$ with $\Psi_0 = I_2$, B is the lower-triangular Cholesky factor of Σ_u, and e_1 selects the first column of the structural impact matrix, isolating the propagation of the first shock while discarding contemporaneous variation from the second. Because the baseline DGP is linear and covariance stationary, θ_h coincides with the population conditional expectation of the system given the structural shock and constitutes the invariant benchmark that both the iterated VAR(q) specifications and the direct LP(4) regressions attempt to recover in finite samples.

We draw Gaussian innovations throughout, so that the conditional-expectation impulse response coincides with the linear projection that both estimators target (Plagborg-Møller & Wolf, 2021). The choice is otherwise innocuous, since the performance metrics depend only on the second moments of the data, and holding the distribution fixed across all designs ensures that later differences reflect the misspecification channel alone.

Sample sizes are set to $T \in \{100, 250, 500\}$, each generated after a 200-period burn-in to remove initialization effects. The lower bound $T = 100$ matches the median sample size documented by Herbst and Johannsen (2024) in the applied LP literature and also marks a practical floor, since estimating models up to VAR($p + H - 1$) at smaller T rapidly exhausts degrees of freedom. The intermediate $T = 250$ represents a comfortably large empirical dataset, roughly twenty years of monthly or sixty years of quarterly data, while $T = 500$ serves as a large-sample reference for convergence toward the population equivalence of Plagborg-Møller and Wolf (2021).

We compare two classes of models. The first is a fixed LP(4) estimator, which serves as the reference point. At each horizon $h = 1, \ldots, H$, the LP(4) estimator is obtained from the OLS regression

$$y_{1,t+h} = \mu_h + \beta_h \hat{\varepsilon}_{1,t} + \sum_{\ell=1}^{4} \delta'_{h,\ell} y_{t-\ell} + \xi_{h,t}, \quad h = 1, \ldots, 20 \tag{24}$$

where $\hat{\varepsilon}_{1,t} = e_1'(\hat{B}^{(4)})^{-1}\hat{u}_t^{(4)}$ is the estimated structural shock recovered from a first-stage VAR(4). While the true structural shock is perfectly observable by construction in our data-generating process, directly projecting on this true innovation would artificially advantage the local projection by bypassing identification uncertainty. To maintain a rigorously fair comparison, we intentionally restrict the LP(4) to use the empirically estimated shock. This guarantees that both estimators face the exact same identification problem. Consequently, any divergence in finite-sample performance is driven strictly by differences in dynamic propagation rather than discrepancies in shock recovery. The structural impulse response estimate at horizon h is then simply the slope coefficient $\hat{\theta}_h^{\text{LP}} = \hat{\beta}_h$.

The second class consists of a sweep of VAR estimators of increasing order, VAR(q) for $q = 1, \ldots, p + H - 1$, where $p = 4$ and $H = 20$. For each order q, the VAR is estimated equation-by-equation via OLS. The reduced-form covariance matrix $\hat{\Sigma}_u^{(q)}$ is Cholesky-decomposed to recover the structural impact matrix $\hat{B}^{(q)}$, and the h-step ahead structural IRF is computed as

$$\hat{\theta}_h^{\mathrm{VAR}(q)} = \hat{\Psi}_h^{(q)} \hat{B}^{(q)} e_1 \tag{25}$$

where $\hat{\Psi}_h^{(q)}$ denotes the h-step ahead moving-average matrix of the estimated VAR(q).

No data-dependent lag selection is performed within the sweep. Every VAR order $q = 1, \ldots, p + H - 1$ is estimated on each simulated sample, so the comparison evaluates fixed specifications rather than data-driven selection procedures.

Performance is assessed along two dimensions, both reported horizon by horizon for $h = 1, \ldots, 20$. For point estimation we evaluate the bias, variance, mean squared error, and root mean squared error of the impulse response estimates relative to the true population estimand, computed across $B = 5{,}000$ Monte Carlo replications (Li et al., 2024). Monte Carlo standard errors accompany every point-estimation measure to quantify the remaining simulation uncertainty.

For inference we assess the empirical coverage of nominal 95% confidence intervals, the fraction of replications whose interval brackets the true structural estimand. For the VAR(q) sweep these intervals are built by the delta method, which propagates the reduced-form coefficient uncertainty through the nonlinear mapping to the structural responses (Kilian & Lütkepohl, 2017). For the fixed LP(4) they use heteroskedasticity-robust standard errors, which are asymptotically valid here given the martingale-difference property of the structural shock under recursive identification (Plagborg-Møller & Wolf, 2021).

3.2 Simulation Results

Figure 2 traces the horizon-averaged root mean squared error against VAR order q for the three persistence regimes and sample sizes, with LP(4) as a horizontal reference. Two level effects set the stage. The error falls monotonically with the sample size and rises with persistence, so the whole frontier and its LP(4) line shift down as T grows and up as ρ grows. At the conventional equal-lag point $q = 4$, the VAR sits below its LP(4) line in every panel, representing the familiar reading that low-order VARs are the more efficient point estimators.

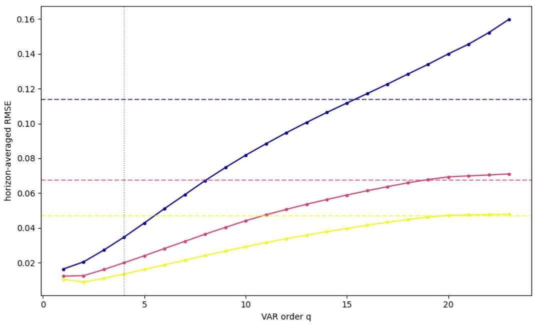
The frontier shows exactly where LP(4) sits once the equal-lag restriction is dropped. The optimal model is a low VAR order, passing a shallow minimum near $q = 2$, before the curve rises into a U-shape that sharpens with persistence ranging from a smooth curve at $\rho = 0.5$ to a steep drop and a pronounced U-shape near the unit-root boundary. At the smallest sample size ($T = 100$), the rising frontier crosses the LP(4) line at $q = 15$ for $\rho = 0.5$, and at $q = 14$ for both $\rho = 0.7$ and $\rho = 0.95$. Thus, even in small samples, the fixed LP(4) already carries the effective complexity of a heavily parameterized VAR rather than a parsimonious one.

The sample size sharpens this picture. As T grows, the frontier flattens and its minimum drifts to higher orders. At $T = 250$, the equivalence point shifts rightward to $q = 19$, $q = 18$, and approximately $q = 17$ across the three respective persistence regimes. At $T = 500$, the convergence is almost complete, with intersections reaching $q = 20$ for both $\rho = 0.5$ and $\rho = 0.7$, and $q = 19$ for $\rho = 0.95$.

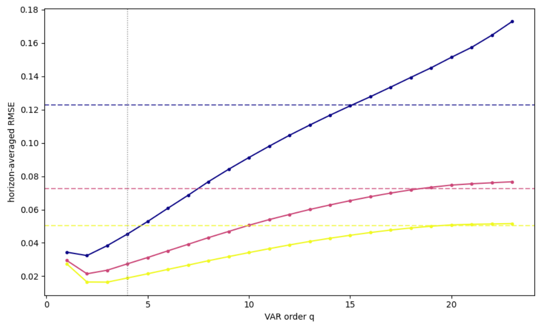
Figure 3 reports bias, variance, MSE, and RMSE at $T = 250$, horizon by horizon across the three persistence regimes, with LP(4) against a representative VAR set. Bias and variance are ordered by lag length, the classic tradeoff. The low-order VARs are the most biased but shed it fastest, and higher orders track the LP(4) bias path progressively longer, with VAR(23) following it throughout. In variance the ordering reverses, the low-order VARs collapsing toward zero earliest while the heavily parameterized VAR(16) and VAR(23) overshoot LP(4) before matching it later. LP(4) itself stays the stable middle reference, and root mean squared error follows variance apart from a short-horizon VAR(2) spike that carries its bias.

Figure 2: Complexity Frontier Across Persistence Regimes

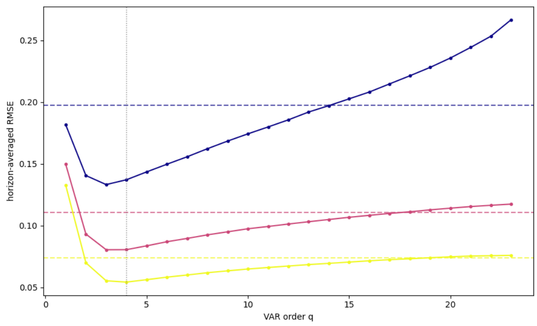
(a) $\rho = 0.5$



(b) $\rho = 0.7$



(c) $\rho = 0.95$



Note. Horizon-averaged RMSE of the structural IRF estimate as a function of VAR order $q = 1, \ldots, 23$ (solid lines), with the fixed LP(4) estimator shown as a horizontal reference line of the same colour (dashed), for each sample size $T \in \{100, 250, 500\}$. The vertical dotted line marks the equal-lag VAR(4), and $B = 5{,}000$ replications are used. Panels correspond to the persistence regimes $\rho \in \{0.5, 0.7, 0.95\}$. The colours blue, red, and yellow represent sample sizes 100, 250, and 500, respectively.

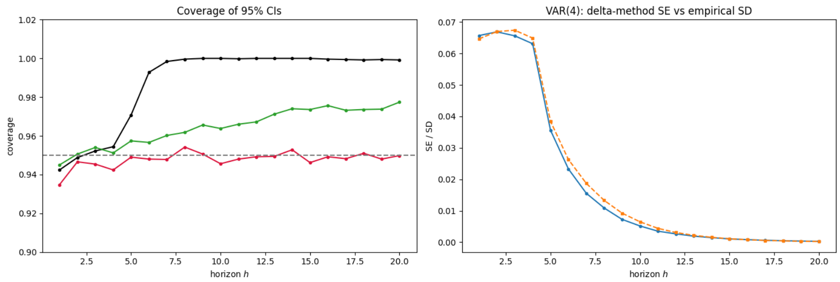
Persistence scales this pattern rather than changing it. The horizon at which the estimators diverge moves slightly later as ρ rises, and the levels grow. At $\rho = 0.7$ the same ordering holds, with VAR(2) now overshooting enough to leave the correctly specified VAR(4) the most accurate. Near the unit-root boundary at $\rho = 0.95$ everything is elevated, VAR(2) turns erratic, the remaining VARs and LP(4) share a common downward path, and because none regains efficiency at long horizons the root mean squared error takes a distinctly arched profile.

Figure 4 reports the empirical coverage of nominal 95 percent confidence intervals at $T = 250$ across the three persistence regimes, alongside the VAR(4) delta-method standard error against its empirical sampling standard deviation. The LP(4) holds near nominal throughout, while the VAR(4) delta-method intervals are strongly persistence-sensitive, over-covering at low persistence and under-covering badly near the unit root.

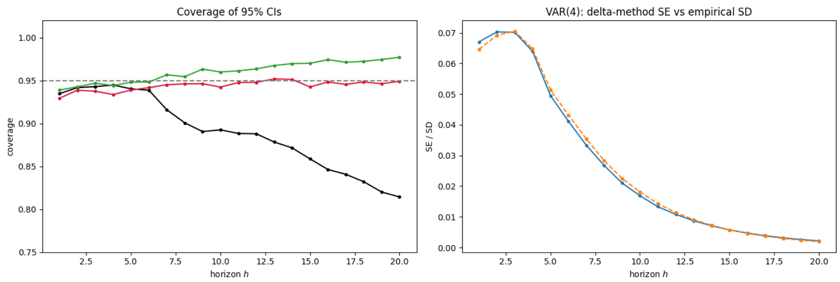
Persistence drives the pattern. The LP(4) stays tight around 0.95 at low and intermediate persistence and slips only to about 0.90 at $\rho = 0.95$. The VAR(4) instead swings with persistence, climbing to full coverage at $\rho = 0.5$ and then deteriorating until it falls below 0.80 at $\rho = 0.95$, the worst of the three. The high-order VAR(23) tracks the same arc more gently, drifting above

Figure 4: Confidence-Interval Coverage Across Persistence Regimes

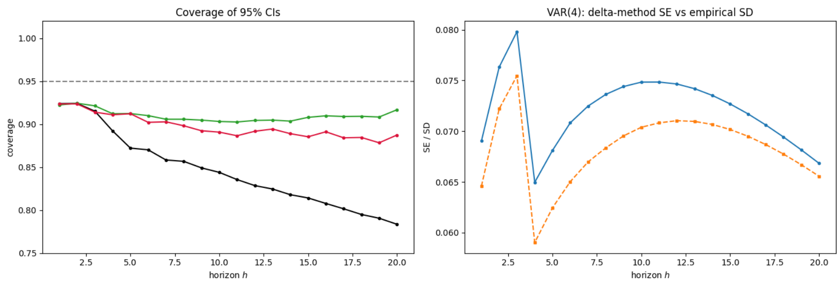
(a) $\rho = 0.5$



(b) $\rho = 0.7$



(c) $\rho = 0.95$



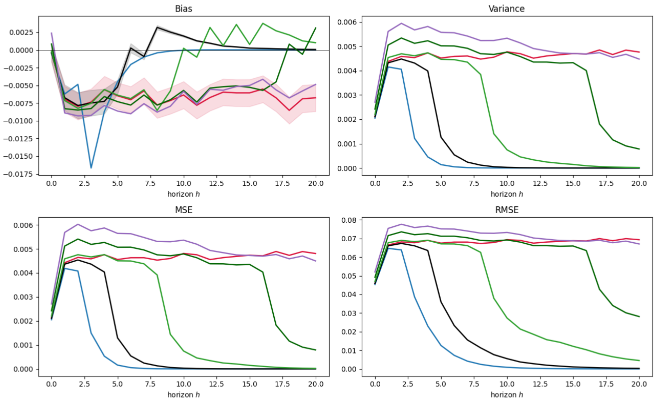
Note. Each subfigure corresponds to a persistence regime $\rho \in \{0.5, 0.7, 0.95\}$ and contains two panels. The *left panel* reports the empirical coverage of nominal 95% confidence intervals, the fraction of replications in which the interval contains the true estimand θ_h, horizon by horizon for $h = 1, \ldots, 20$, with $T = 250$ and $B = 5{,}000$. Coverage is shown for VAR(4) (black), VAR(23) (green), and LP(4) (red), with the dotted horizontal line marking the nominal 95% level. VAR(q) intervals are constructed via the delta method, and LP(4) intervals use heteroskedasticity-robust standard errors. The *right panel* validates the VAR(4) delta-method standard error against the empirical Monte Carlo sampling standard deviation of $\hat{\theta}_h$. The empirical SD is shown in blue and the mean delta-method SE in orange, so that close agreement indicates an accurate analytical SE while a gap signals that it misstates the true sampling dispersion.

nominal at low persistence and settling near the LP(4) around 0.90 at the boundary.

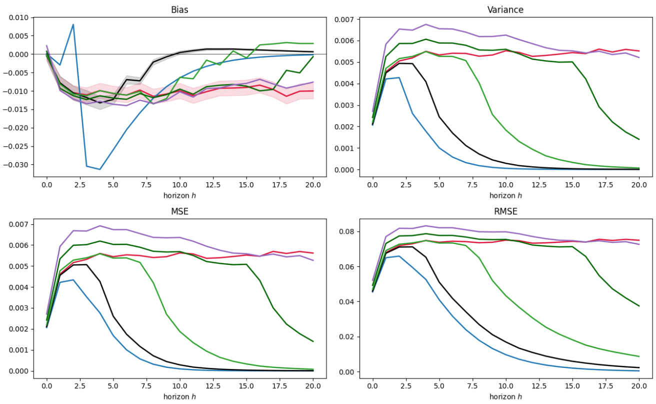
The right column shows the miscalibration is not a broken standard error. At $\rho = 0.5$ and

Figure 3: Point-Estimation Performance Across Persistence Regimes

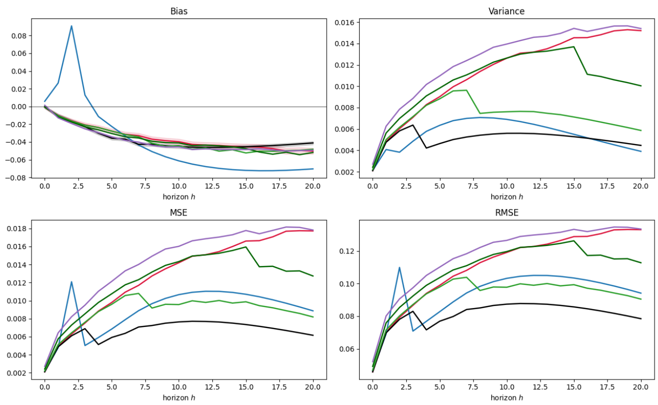
(a) $\rho = 0.5$



(b) $\rho = 0.7$



(c) $\rho = 0.95$



Note. Each panel reports bias, variance, and RMSE of the estimated impulse responses $\hat{\theta}_h$ relative to the true estimand θ_h, horizon by horizon for $h = 1, \ldots, 20$, computed across $B =$ 5,000 Monte Carlo replications with $T = 250$. The fixed LP(4) reference line is compared against the complexity-adjusted VAR(q) sweep. Panels correspond to the persistence regimes $\rho \in \{0.5, 0.7, 0.95\}$. The shaded area in the bias panel is the 95% band from the Monte Carlo standard errors. Line colour denotes the model, where blue, black, light green, dark green, lilac, and red represent VAR(2), VAR(4), VAR(8), VAR(16), VAR(23), and LP(4), respectively.

$\rho = 0.7$ the VAR(4) delta-method standard error and the empirical sampling standard deviation are nearly indistinguishable, so the coverage gaps there are structural. Only at $\rho = 0.95$ does a vertical gap open, with the empirical dispersion sitting above the analytical standard error at short and intermediate horizons before the two converge by the end of the window.

3.3 Discussion of Results

Because an LP(p) algebraically embeds a sequence of VAR predictions of orders p through $p + h - 1$, its coherent benchmark is the full VAR-order sweep rather than a single equal-order VAR. The point-estimation advantage of the equal-lag VAR(4) replicates the standard consensus (Li et al., 2024; Montiel Olea et al., 2025), but locating the local projection on the frontier reveals where it actually falls. The LP(4) does not sit near the parsimonious end. It behaves like a highly parameterized system, intersecting the sweep beyond $q \approx 15$.

The bias–variance tradeoff is governed by persistence. Under low persistence the frontier is monotonic, so the variance penalty for over-parameterization outweighs any bias reduction once the impulse response decays quickly. Near the unit-root boundary the pronounced U-shape instead exposes the cost of lag truncation, where the downward bias in the autoregressive coefficients dominates the error budget (Kilian, 1998). Here the local projection loses its robustness advantage. It provides no relief from this persistence-driven bias while carrying the higher variance of its horizon-by-horizon regressions, so it is dominated by the equal-lag VAR.

Root mean squared error alone obscures a divergence between point estimation and inference. The coverage results show that the VAR's parametric efficiency comes at the cost of inferential fragility, as the VAR(4) delta-method intervals fail for structural reasons across the persistence spectrum. At weak persistence they over-cover, because the response converges to zero and the interval captures the unconditional mean rather than signalling precision. At high persistence the first-order Taylor approximation behind the delta method breaks down and understates the dispersion of the skewed finite-sample distribution (Kilian, 1999), producing steep under-coverage. The LP(4), by contrast, holds near-nominal coverage across all regimes, with its heteroskedasticity-robust standard errors remaining valid even as the process approaches non-stationarity (Montiel Olea & Plagborg-Møller, 2021). Crucially, however, this inferential fragility is largely a consequence of the strict parsimony imposed by the equal-lag comparison. When evaluating the full complexity frontier, the finite-sample gap between the two estimator classes systematically closes. A sufficiently over-parameterized model, such as the VAR(23), mimics the flexible behavior of the direct projection, matching its bias reduction and nearly replicating its robust coverage levels even at high persistence.

4. Extension 2: Functional Misspecification

The baseline established a clear reference under correct specification, where the equal-lag VAR is the more efficient point estimator and the local projection offers more reliable inference. This extension relaxes correct specification through the conditional mean rather than the dynamics, augmenting the linear VAR with a nonlinear term so that the fitted models become linear projections of a nonlinear process. The motivating question is whether the local projection gains a genuine finite-sample advantage once the linear model is only an approximation to the truth, and where any such advantage appears, in point estimation or in inference.

4.1 Simulation Design

To relax correct specification in the conditional mean, the baseline VAR(4) is augmented with a mean-zero quadratic term in the first lag, yielding a nonlinear DGP:

$$y_t = A_1 y_{t-1} + A_2 y_{t-2} + A_3 y_{t-3} + A_4 y_{t-4} + \gamma \cdot g(y_{1,t-1}) + u_t, \quad u_t = B\varepsilon_t, \quad \varepsilon_t \overset{\text{iid}}{\sim} \mathcal{N}(0, I_2) \tag{26}$$

where

$$g(y_{1,t-1}) = \left(0,\ \left(\frac{y_{1,t-1}}{\sigma_1}\right)^2 - 1\right)' \tag{27}$$

is a standardised, mean-zero quadratic in the first variable's first lag, entering the second equation only, with $\sigma_1 = \sqrt{\mathrm{Var}(y_{1,t})}$ the stationary standard deviation of $y_{1,t}$ under the linear ($\gamma = 0$) VAR. The autoregressive matrices and the impact matrix B are inherited from the baseline, so the nonlinear term is the only departure.

The nonlinearity enters the propagation through a quadratic in $y_{1,t-1}$ while the structural shocks still enter linearly through $u_t = B\varepsilon_t$, so the design probes functional-form misspecification of the conditional mean rather than nonlinear shock transmission. Because the offending term is a function of lagged outcomes, lengthening a linear VAR cannot remove it, and every fitted model remains a linear projection of a nonlinear process. The quadratic also raises the risk of explosive paths at high persistence, so persistence is restricted to $\rho \in \{0.5, 0.7\}$ to keep the process covariance stationary and the estimand well-defined.

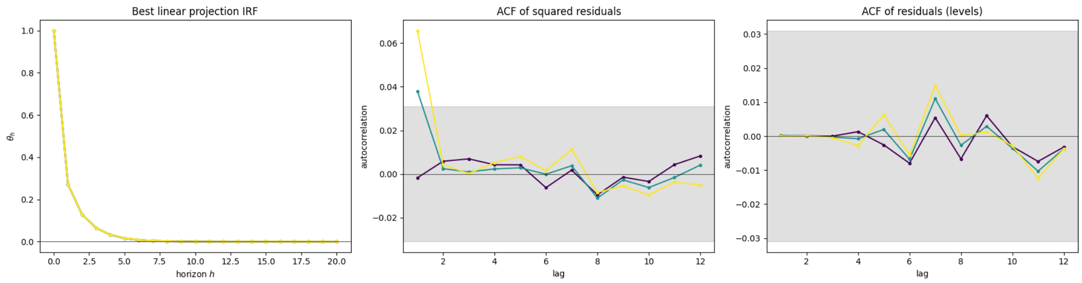
The estimand is the best linear approximation to the nonlinear impulse response, $\theta_h^{\text{lin}} = \Psi_h^{\text{lin}} B^{\text{lin}} e_1$, where Ψ_h^{lin} is the linear-projection response implied by the covariance structure of the nonlinear process and B^{lin} the corresponding impact matrix under the baseline recursive identification. It is well-defined because Plagborg-Møller and Wolf (2021) establish that LP and VAR target the same best linear approximation in population whether or not the DGP is linear (Proposition 1). Both estimators therefore aim at the same object, and any finite-sample divergence reflects bias and variance rather than differential robustness to the nonlinearity.

Persistence is varied across $\rho \in \{0.5, 0.7\}$ and the nonlinearity strength across $\gamma \in \{0.3, 0.6\}$, with $\gamma = 0$ recovering the linear baseline, jointly with two sample sizes $T \in \{250, 10{,}000\}$, the first a larger-horizon macro sample and the second approximating the asymptotic limit. Both estimators are applied without modification, so the functional-form misspecification is absorbed entirely into their finite-sample performance. The performance measures match the baseline, augmented by a small-sample and asymptotic comparison of each estimator's deviation from the true linear-projection IRF, which directly tests the population-equivalence claim.

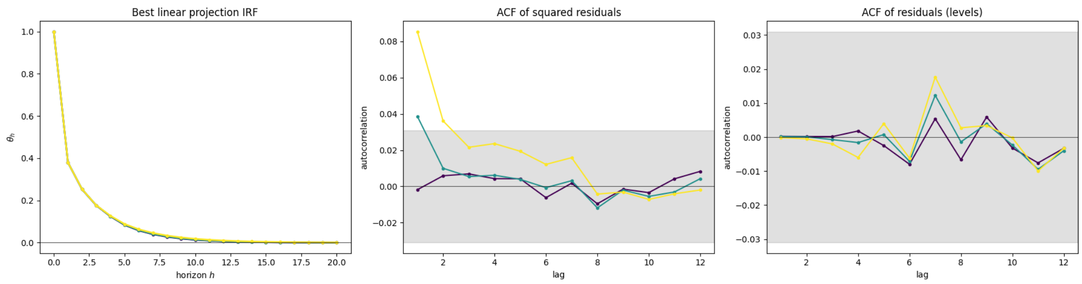
Figure 5 fixes the two facts that define this design. The left panel shows the best linear-projection IRF is almost invariant to γ, because the centered quadratic is approximately orthogonal to the linear lag space and leaves the linear projection that both estimators target essentially unchanged. The residual diagnostics show where the nonlinearity does bite. The autocorrelation of the squared second-equation residual lifts above the white-noise band as γ grows, the signature of conditional heteroskedasticity, while the level residual stays white, so the conditional mean is captured and only the variance moves with γ. Both estimators therefore aim at the same near-constant target, and the design isolates how each copes with the heteroskedasticity rather than with any movement in the estimand.

Figure 5: True IRF Functions by Nonlinearity Strength

(a) $\rho = 0.5$



(b) $\rho = 0.7$



Note. Each subfigure ($\rho \in \{0.5, 0.7\}$) has three panels. *Left*, the best linear-projection IRF θ_h of variable 1 to shock 1, approximately invariant across nonlinearity strength γ. *Middle and right*, the ACF of the squared and the level second-equation residual ($\hat{e}_{2,t}^2$ and $\hat{e}_{2,t}$). The squared-residual ACF lifts above the band as γ grows, the signature of conditional heteroskedasticity, while the level residual stays white. Lines are purple $\gamma = 0$ (baseline), teal $\gamma = 0.3$, and yellow $\gamma = 0.6$, and the grey area is the $\pm 1.96/\sqrt{N}$ 95% white-noise band.

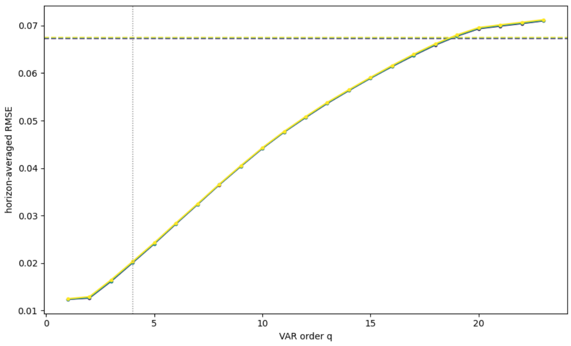
4.2 Simulation Results

Figure 6 traces the frontier at a small sample ($T = 250$) and near the asymptotic limit ($T = 10{,}000$). In small samples the LP(4) sits high on the frontier, with an RMSE matching a VAR of order 18 to 19 at both persistence levels, while the parsimonious VAR(1) and VAR(2) are among the lowest-RMSE specifications because their low variance masks an underlying lag-truncation bias. The nonlinearity barely registers here, leaving the frontier unchanged at $\rho = 0.5$ and lifting it only mildly at $\rho = 0.7$, since the misspecification needs persistence to bite.

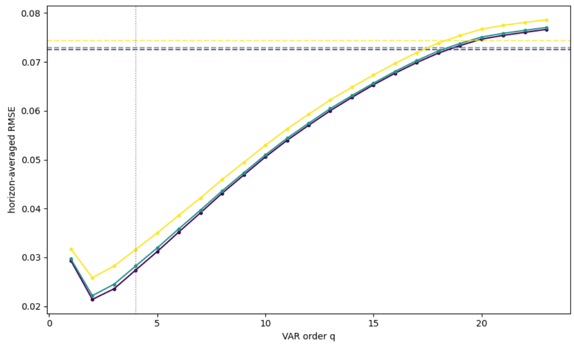
The asymptotic frontier exposes what the small-sample variance hid. With the variance gone, the under-lagged VAR(1) and VAR(2) blow up, as they reveal a different target linear projection, while VAR(3) and VAR(4) become the lowest-RMSE specifications and the rest of the sweep collapses onto LP(4) at both persistence levels. That collapse is the population equivalence made visible, with every adequately lagged model recovering the same linear projection. The remaining γ effect shrinks with the sample size, confirming that the nonlinearity acts through an inflated, asymptotically vanishing variance rather than through bias.

Figure 6: Complexity Frontier under Nonlinear Lag Misspecification

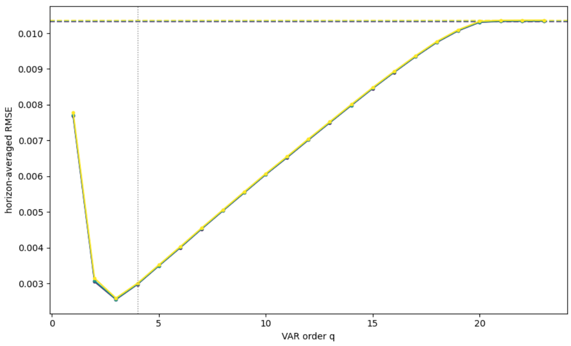
(a) $\rho = 0.5$ (Sample)



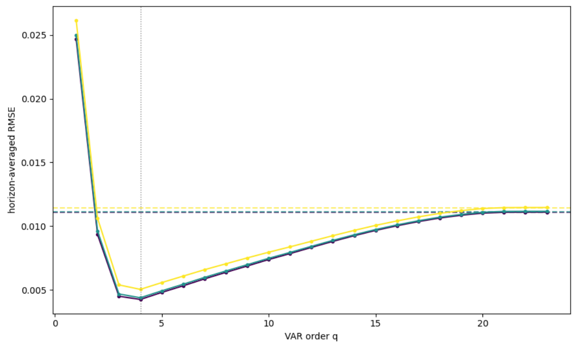
(b) $\rho = 0.7$ (Sample)



(c) $\rho = 0.5$ (Asymptotic)



(d) $\rho = 0.7$ (Asymptotic)



Note. Horizon-averaged RMSE of the structural IRF against VAR order $q = 1, \dots, 23$ (solid), with LP(4) a same-colour horizontal dashed reference and the vertical dotted line marking the equal-lag VAR(4). There is one colour per nonlinearity strength, with purple $\gamma = 0$ (baseline), teal $\gamma = 0.3$, and yellow $\gamma = 0.6$. The top row (a, b) is the small-sample frontier $T = 250$ and the bottom row (c, d) the asymptotic frontier $T = 10{,}000$. Within each row $\rho \in \{0.5, 0.7\}$.

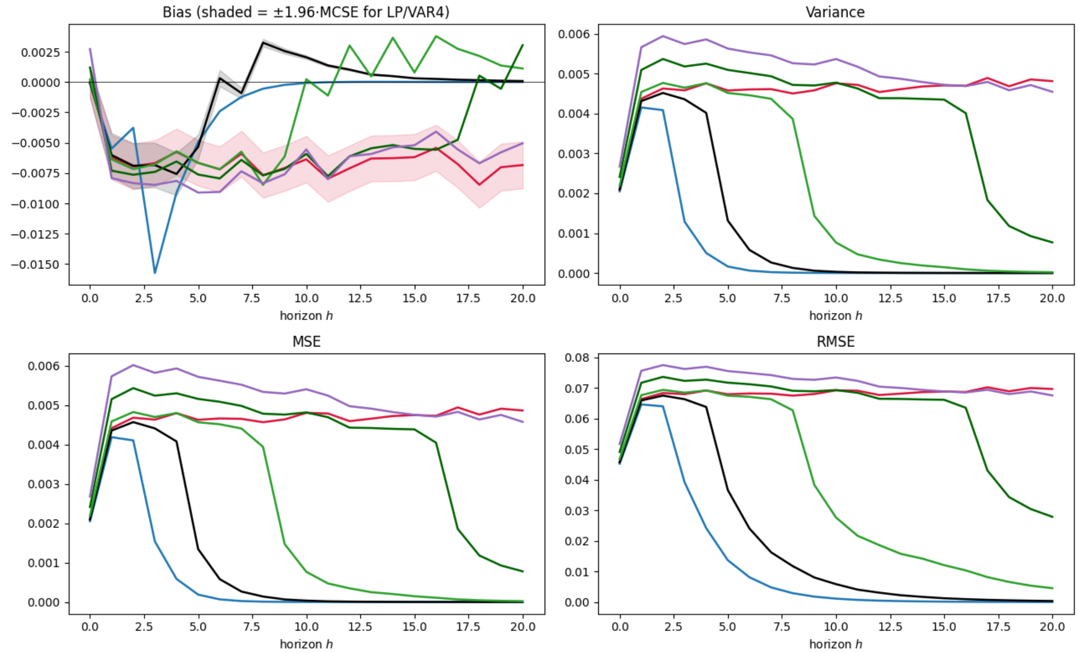
The point-estimation figures (Figures 7 and 8) sharpen the equivalence further. In small samples the bias is noisy, with each VAR tracking the LP(4) bias up to its own lag order and the LP(4) holding a small, roughly constant negative bias. Asymptotically every model except the under-lagged VAR(2) collapses onto zero bias at both persistence levels. VAR(2) alone keeps a visible oscillating bias, peaking near 0.01 at $\rho = 0.5$ and 0.025 at $\rho = 0.7$ before decaying, because its insufficient lag length targets a different linear projection than the one LP(4) and the higher-order VARs share.

Variance tells the complementary story. Each VAR traces the LP(4) variance and then turns down about one horizon before its lag order, so the high-order VAR(23) never turns within the $h \leq 20$ window, while the LP(4) variance stays roughly flat. Higher persistence keeps the response, and hence the variance, elevated for longer, and asymptotically the curves trace LP(4) almost exactly before converging to zero. Root mean squared error mirrors variance, the one exception being VAR(2), which spikes at short horizons. Because that spike shows up in bias and root mean squared error but not in variance, it is a wrong-estimand effect of insufficient lag length rather than a variance failure.

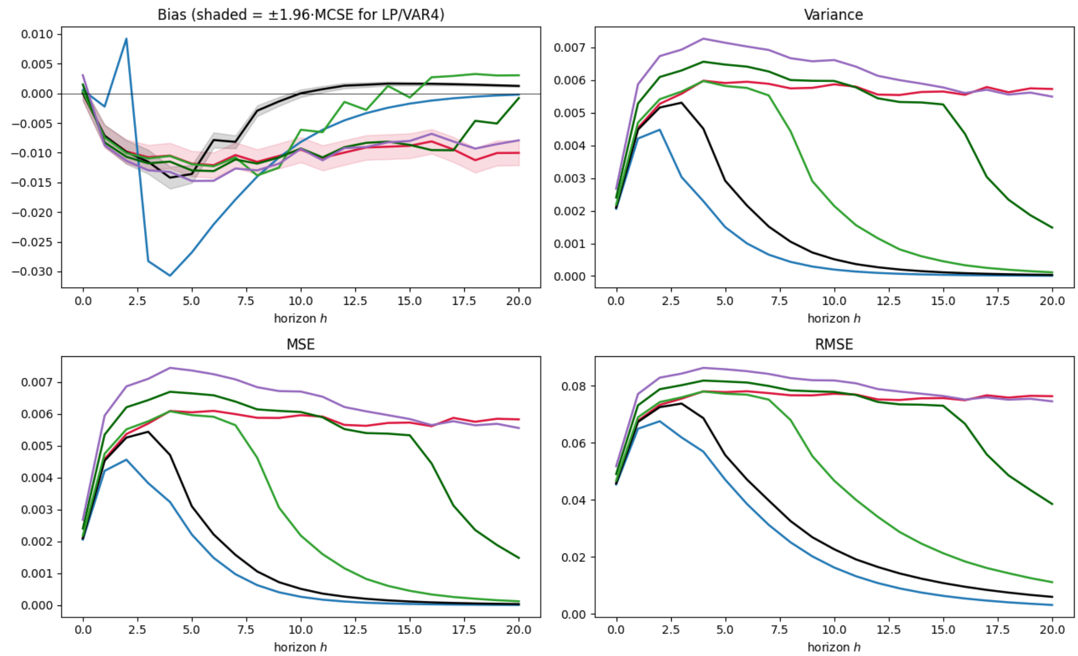
The staggered turning points are an artifact of iteration. A VAR(q) builds its horizon-h response by iterating the estimated companion, so while $h \leq q$ the response still draws directly on the estimated lag matrices and both its bias and variance stay elevated, but once $h > q$ each further step multiplies by a contraction and both damp geometrically. Higher-order VARs therefore turn down later. The LP(4) never iterates, estimating each horizon by a separate regression, so its bias and variance stay flat across horizons. The turning points reflect the iteration mechanism, not a decaying estimand or any genuine improvement of the VAR at long horizons.

Figure 7: Sample Point Estimates under Strong Nonlinearity

(a) $\rho = 0.5$



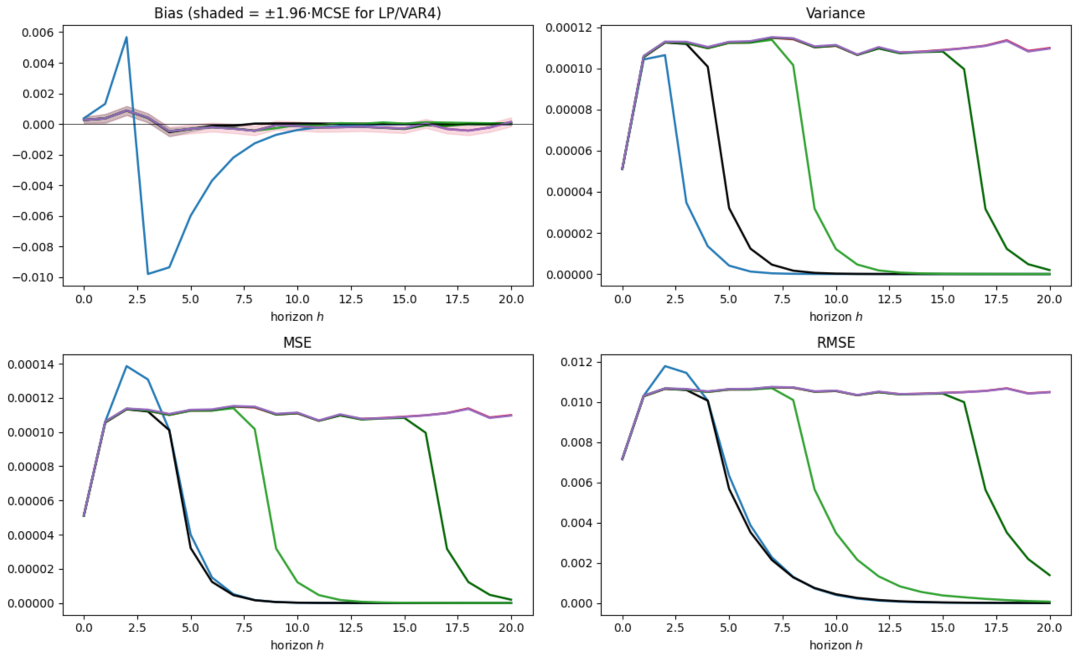
(b) $\rho = 0.7$



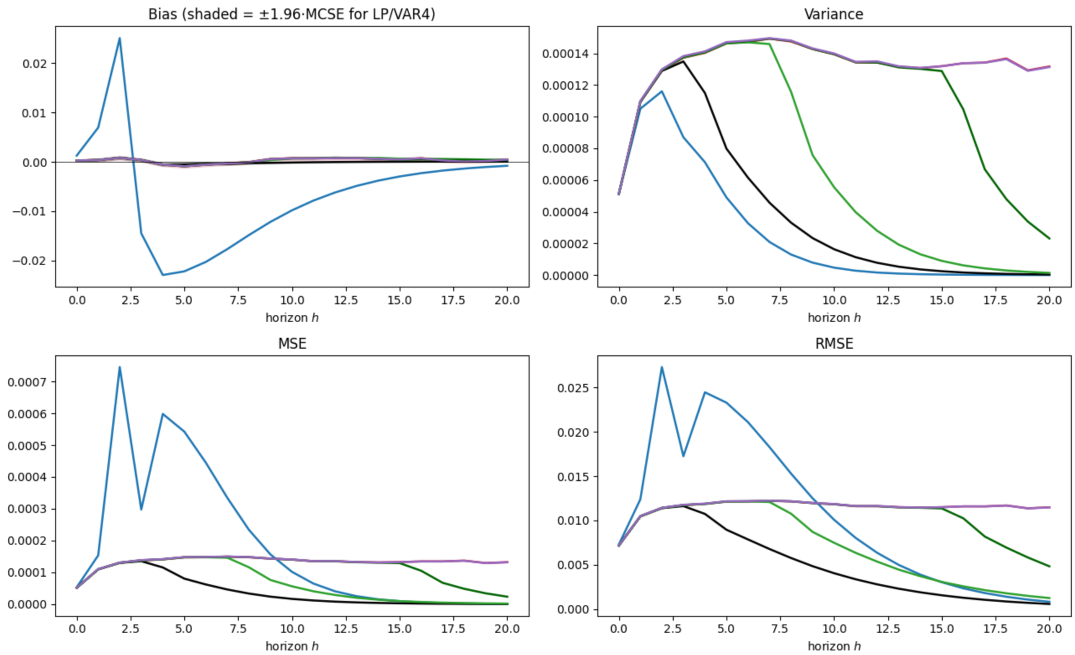
Note. Bias, variance, MSE, and RMSE of $\hat{\theta}_h$ relative to the true (linear-projection) estimand θ_h for the nonlinear lag-quadratic DGP, at $T = 250$, $B = 5{,}000$, $\gamma = 0.6$. LP(4) (red) is compared against a representative VAR(q) set, namely VAR(2) blue, VAR(4) black, VAR(8) light green, VAR(16) dark green, and VAR(23) purple. Rows correspond to $\rho \in \{0.5, 0.7\}$. The shaded band on bias is the 95% Monte Carlo standard-error band.

Figure 8: Asymptotic Point Estimates under Strong Nonlinearity

(a) $\rho = 0.5$



(b) $\rho = 0.7$



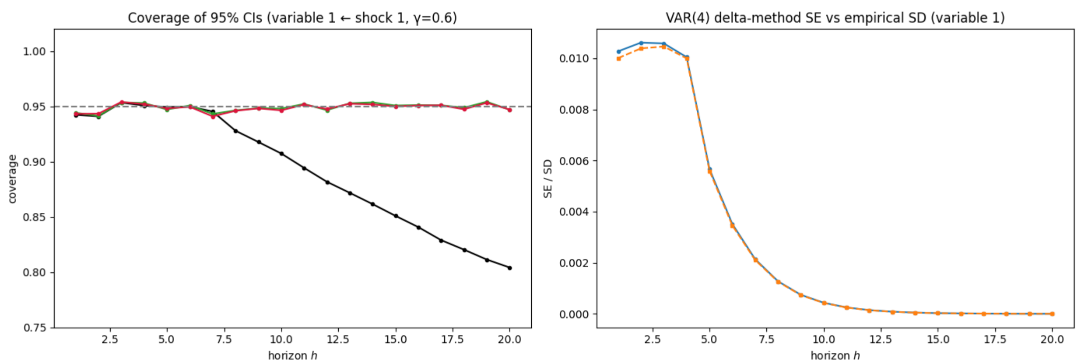
Note. As in the sample point-estimation figure but at $T = 10{,}000$ (approximating the asymptotic limit), $B = 5{,}000$, $\gamma = 0.6$. The estimators are LP(4) red, VAR(2) blue, VAR(4) black, VAR(8) light green, VAR(16) dark green, and VAR(23) purple. Rows correspond to $\rho \in \{0.5, 0.7\}$. The shaded band on bias is the 95% Monte Carlo standard-error band.

Figure 9 reports coverage at the strongest nonlinearity $\gamma = 0.6$ for variable 1, alongside the VAR(4) delta-method standard error against its empirical sampling standard deviation. The heteroskedasticity splits the estimators as expected. The LP(4), with heteroskedasticity-robust standard errors, holds near the nominal 95% level across both persistence regimes, while the VAR(4), whose delta-method intervals assume homoskedasticity, under-covers, sliding below nominal after the short horizons and worsening from $\rho = 0.5$ to $\rho = 0.7$ as the more variable state amplifies the heteroskedasticity. The high-order VAR(23) sits between the two, since its richer parameterization absorbs more of the dynamics.

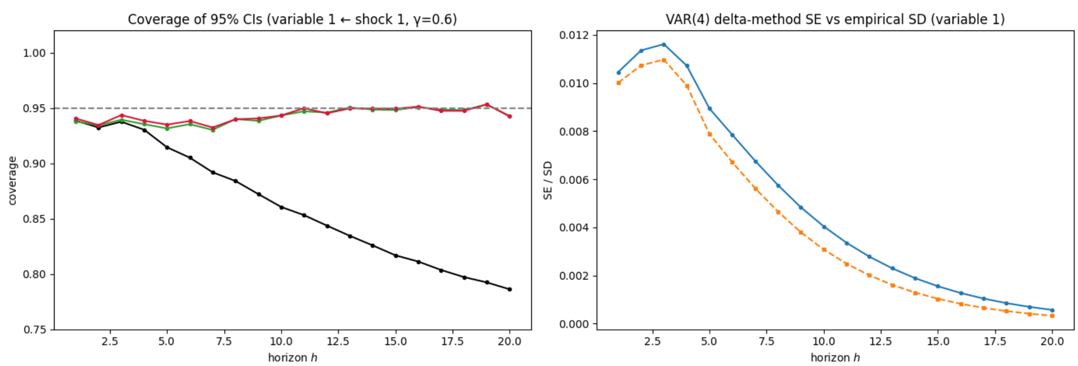
The right panel shows why the VAR(4) fails. Its delta-method standard error sits visibly below the empirical sampling standard deviation, so the intervals are too narrow. This is the conditional heteroskedasticity from the design figure surfacing in inference, as the homoskedastic delta method cannot see the variance that scales with $y_{1,t-1}^2$, whereas the heteroskedasticity-robust standard errors of LP(4) absorb it. That is why only the local projection stays calibrated under the nonlinearity.

Figure 9: Confidence-Interval Coverage under Nonlinear Lag Misspecification

(a) $\rho = 0.5$



(b) $\rho = 0.7$



Note. Empirical coverage of nominal 95% CIs by horizon ($h = 1, \ldots, 20$) for the response of variable 1 to shock 1 under the nonlinear lag-quadratic DGP at the strongest nonlinearity $\gamma = 0.6$, $T = 10{,}000$, $B = 5{,}000$. In each subfigure the *left* panel reports coverage of VAR(4) (black), VAR(23) (green), and LP(4) (crimson), with the dashed horizontal line at the nominal 95% level. The *right* panel validates the VAR(4) delta-method SE against the empirical Monte Carlo sampling SD of $\hat{\theta}_h$. VAR(q) intervals use the delta method, and LP(4) uses heteroskedasticity-robust SEs. Subfigures span $\rho \in \{0.5, 0.7\}$.

4.3 Discussion of Results

The results confirm that the population equivalence of Plagborg-Møller and Wolf (2021) survives this functional misspecification. Because the centred lag-quadratic is approximately orthogonal to the linear lag space, it leaves the best linear projection essentially unchanged, so the estimand is nearly γ-invariant and both LP(4) and the adequately lagged VARs recover it asymptotically with no bias floor. The only systematic departures are the under-lagged VAR(1) and VAR(2), which target a different linear projection and keep a non-vanishing bias. That is not a violation of the equivalence but its precondition, since the result holds only at sufficient lag length.

The gap between this population equivalence and the finite-sample divergence is exactly what the equivalence asserts. Plagborg-Møller and Wolf (2021) equate the two estimands, the probability limits the procedures converge to, not the estimates themselves. Both are functions of the same population second moments, and at those true moments the direct projection of the local projection and the iterated coefficients of the VAR are two algebraic routes to the same impulse response. In a finite sample the moments are replaced by noisy estimates, and the two routes propagate that noise differently. The VAR compounds it through the iterated companion while the local projection estimates each horizon by a separate regression, and because the impulse response is a nonlinear function of the coefficients each route also carries its own finite-sample bias. The estimators therefore share a target but not a sampling distribution, and they coincide only in the limit.

That shared target but separate sampling distribution splits the estimators along the two dimensions we measure. In point estimation the VAR(4) keeps the mean squared error advantage, because both carry the same small bias and the difference is variance, and the direct per-horizon regressions of the local projection absorb more sampling noise than the iterated VAR. In inference the ranking reverses. The lagged-state nonlinearity injects conditional heteroskedasticity into the residuals, which the homoskedastic delta-method intervals cannot accommodate, so the equal-lag VAR(4) under-covers most severely as γ and persistence rise. Higher-order VARs recover part of the way as their richer parameterization absorbs more of the dynamics, while the heteroskedasticity-robust interval of the LP(4) stays near nominal throughout.

These coverage differences are about inference, not the point estimand, so they do not contradict the equivalence. The two estimators still target the same impulse response, and only their interval validity differs once the standard-error assumptions are violated. The upshot is a clean division of labour, the VAR for the efficiency of the point estimate and the local projection for valid inference under the heteroskedasticity.

5. Conclusion

This paper applied the complexity-adjusted benchmark of Ludwig (2026) to the finite-sample comparison of Vector Autoregressions and Local Projections, replacing the conventional equal-lag contest with one that holds effective complexity constant. Under correct specification a fixed $LP(4)$ already behaves like a highly parameterized $VAR(q \approx 15)$, so the familiar point-estimation edge of low-order VARs is largely an artifact of parsimony rather than an intrinsic advantage of the iterated method. Functional misspecification, introduced through a nonlinear lag-quadratic process, leaves this reading intact. Because the centered quadratic is approximately orthogonal to the linear lag space, the estimand stays nearly invariant to the nonlinearity, and asymptotically both $LP(4)$ and the adequately lagged VARs collapse onto the same best linear approximation, so the population equivalence of Plagborg-Møller and Wolf (2021) survives without an asymptotic bias floor. In finite samples the $VAR(4)$ keeps its mean squared error dominance, since the direct horizon-by-horizon regressions of the local projection carry a larger variance.

The advantage that is absolute is inferential. The unmodeled quadratic injects conditional heteroskedasticity into the residuals, which the homoskedastic delta-method intervals cannot capture, so they understate the sampling dispersion and under-cover ever more severely as persistence and nonlinearity rise. The failure is sharpest for the equal-lag $VAR(4)$ and eases along the complexity frontier, since the richer parameterization of higher-order VARs absorbs more of the dynamics and pulls their coverage back toward the near-nominal level that the heteroskedasticity-robust intervals of $LP(4)$ hold throughout. This is the first of our designs in which the robustness of local projections becomes a tangible advantage, though it is confined to inference and narrows against sufficiently high-order VARs. The broader lesson is that estimand equivalence in population does not imply inferential equivalence in finite samples, and the choice between the two methods is objective-dependent, weighing the efficiency of the VAR point estimate against the interval validity of the direct projection.

Three limitations bound these conclusions. First, the full VAR sweep up to $VAR(23)$ makes smaller samples infeasible, since the high-order systems turn rank-deficient and unstable, leaving the small-sample region, where the bias-variance trade-off is sharpest and an LP advantage most plausible, unexplored. Second, the nonlinear lag-quadratic destabilizes the simulated paths at high persistence, so we restricted persistence to $\rho \in \{0.5, 0.7\}$ and left the near-unit-root region unexamined, precisely where the interaction of high persistence and functional misspecification would most strain inference. Third, the VAR coverage failure reflects the default homoskedastic delta-method standard error rather than the iterated architecture itself, and equipping the VAR sweep with robust or wild-bootstrap standard errors would likely narrow the inferential gap.

Future work should extend the complexity-adjusted frontier to nonlinear settings that permit high persistence without explosive paths, so the near-unit-root region can be revisited, and should formalize a complexity adjustment for bootstrap-based inference. The latter would deliver a cleaner horse race, isolating whether the robust coverage of local projections is an intrinsic virtue of direct multi-step forecasting or merely a mechanical benefit of robust variance estimation.

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Appendix

A Code and Data Availability

All code and replication materials for both extensions are available at github.com/Aaron030/Monte_Carlo-LP_vs_VAR. The repository contains the simulation code, the routines that generate every figure, table summaries of the figures, and instructions to reproduce the full study.

Declaration of AI Usage

During the preparation of this seminar paper, I Patrick Tenner, used Claude for an overview of references, early drafting, coding assistance, as well as Claude and Gemini for linguistic smoothing. I have reviewed and edited the content as needed and take full responsbility.

Patrick Tenner/Aaron Liebig